# Worksheet 4

## Exercise 1: Shoe Size

The dataset `d.shoes` in `shoes.rda` contains gender, income and shoe size information for 40 individuals.

```r
# Einlesen der Daten
load("shoes.rda")
```

a) Fit a simple linear regression model to the data, with income (`salary`) as the response variable and shoe size (`shoes`) as the explanatory variable. How do you interpret the result?

```r
fit <- lm(salary ~ shoes, data = d.shoes)
summary(fit)
```

```
Call:
lm(formula = salary ~ shoes, data = d.shoes)

Residuals:
   Min     1Q Median     3Q    Max 
-61.05 -15.72   1.45  13.64  63.64 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -178.415     59.227  -3.012  0.00459 ** 
shoes          7.769      1.492   5.209  6.9e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 27.57 on 38 degrees of freedom
Multiple R-squared:  0.4166,    Adjusted R-squared:  0.4012 
F-statistic: 27.13 on 1 and 38 DF,  p-value: 6.904e-06
```

Shoe size (`shoes`) and income (`salary`) are both influenced by `gender`. This fork structure induces a dependence between shoe size and income when gender is not conditioned on. Consequently, the usual interpretation that an increase of one shoe size leads to a salary increase of 7.8 is not meaningful, as it reflects a spurious association rather than a causal effect.

b) Fit a multiple linear regression model to the data, with income (`salary`) as the response variable and shoe size (`shoes`) as well as gender (`gender`) as the two explanatory variables. Compare the result with subtask a).

```r
fit_adj <- lm(salary ~ shoes + gender, data = d.shoes)
summary(fit_adj)
```

```
Call:
lm(formula = salary ~ shoes + gender, data = d.shoes)

Residuals:
```

```
    Min      1Q  Median      3Q     Max
-52.699 -10.774   0.255  11.954  40.642

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  162.086     76.923   2.107   0.0419 *
shoes         -1.659      2.067  -0.803   0.4273
genderM       65.730     12.080   5.441 3.59e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20.82 on 37 degrees of freedom
Multiple R-squared:  0.6759,	Adjusted R-squared:  0.6584
F-statistic: 38.58 on 2 and 37 DF,  p-value: 8.86e-10
```

Adjusting for `gender` controls for the confounding effect. The coefficient for shoe size can therefore be interpreted as the direct causal effect on income. The estimated effect is statistically not significant, indicating that there is no evidence for a causal effect of shoe size on income in the data.

c) Use the regression models from a) and b) to calculate the expected income for a woman with a shoe size of 38. Explain the difference in the prediction.

```r
x0 <- data.frame(shoes = 38, gender = "F")
predict(fit, newdata = x0)
```

```
       1
116.8191
```

```r
predict(fit_adj, newdata = x0)
```

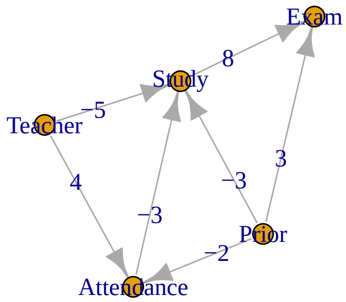
```
       1
99.03975
```

The information about gender is essential for an accurate prediction. In non-causal regression, the expected income is overestimated.

## Exercise 2: Exam Score

In this exercise, we will analyze the effect of study hours on the exam grade (continuation from Worksheet 2). We will introduce a bit more complexity by adding a node for teacher quality and student attendance in class. The following graphical causal model shows all dependencies:

```r
set.seed(253)
library(gRbase)
library(igraph)
grest <- dag(c("Study","Teacher","Prior","Attendance"),
             c("Attendance","Prior","Teacher"),
             c("Exam","Study","Prior"))
E(grest)$weight <- c(-5,-3,-3, -2, 4, 8, 3)
plot(grest, edge.label = E(grest)$weight)
```




Assume that the data generating process is given by linear structural causal model and the numbers in the DAG correspond to the coefficients.

a) What is the direct effect from studying hours on the exam score?

The direct causal effect is given by the weight of the arrow from study to exam. This is 8.

b) What is the total causal effect of class attendance on the exam score?

There is only one directed path from attendance to exam. The total causal effect is the multiplication of weights along the directed path. This is $-3 \cdot 8 = -24$.

c) What is the total causal effect of prior knowledge on the exam score?

There are three directed path from prior to exam:
Prior $\rightarrow$ exam: 3
Prior $\rightarrow$ study $\rightarrow$ exam: $-3 \cdot 8 = -24$
Prior $\rightarrow$ attendance $\rightarrow$ study $\rightarrow$ exam: $-2 \cdot -3 \cdot 8 = 48$
The total causal effect is the sum of the results over all three directed paths:
$3 - 24 + 48 = 27$.

d) Report the R code to estimate the total causal effect of attendance on the exam score.

The Backdoor Criterion gives the following adjustment set:
{Prior, Teacher}
The variables of adjustment set need to be included into the regression:

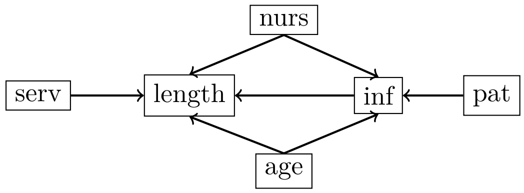
```
fit <- lm(exam ~ attendance + prior + teacher)
coef(fit)["attendance"] # estimated total effect
```

## Exercise 3: Hospital

In a study on the risk of infection in hospitals, the following variables were recorded in a random sample of 112 U.S. hospitals:

`length`: average length of hospitalization of all patients (in days)
`age`: average age of the patients
`inf`: average risk of infection (in percent)
`pat`: average number of patients in hospital per day
`nurs`: proportion of fully employed and educated nurses per patient
`serv`: proportion of services offered per patient

The following graphical model describes the causal relations between these variables:




```r
load("spital.rda")
```

a) Fit a simple linear regression with the response `length` and the explanatory variable `inf`. What does the coefficient of `inf` tell us about the average length of hospitalization?

```r
fit <- lm(length ~ inf, data = spital)
coef(fit)
```

```
(Intercept)         inf
  9.5294016   0.2140981
```

The model shows a significant **association** between the risk of infection and the average length of hospitalization. The association is positive: a higher risk of infection corresponds to longer hospitalizations. From the regression we can't conclude anything about the causal direction, i.e., if we want reduce the risk of infection in a hospital through newly introduced measures, the average length of stay is not necessarily decreasing. Furthermore, the R-squared is only 5.6%. This means that the model explains only 5.6% of the variance of the response.

b) Why does the model in a) not measure the correct total causal effect of inf on length?

The graphical model shows three paths between the two nodes `inf` and `length`. The two nodes `nurs` and `age` are confounders.

c) Determine at least two possible adjustment sets to estimate the true total causal effect of `inf` on `length`?

We need to block the 2 backdoor path between `inf` and `length`:
Path 1: `inf` ← `nurs` → `length`
Path 2: `inf` ← `age` → `length`
Therefore, a possible adjustment set is:
{ `age`, `nurs` }
Further, we can add other non-descendant variables and obtain the following adjustment sets:
{ `age`, `nurs`, `pat` }
{ `age`, `nurs`, `serv` }
{ `age`, `nurs`, `pat`, `serv` }

d) Estimate the total causal effect using at least two different adjustment sets. Do you see differences in the results?

```r
fit_adj1 <- lm(length ~ inf + age + nurs, data = spital)
fit_adj2 <- lm(length ~ inf + age + nurs + pat, data = spital)
fit_adj3 <- lm(length ~ inf + age + nurs + serv, data = spital)
fit_adj4 <- lm(length ~ inf + age + nurs + serv + pat, data = spital)

# Estimated total causal effect
c(adj1 = coef(fit_adj1)["inf"], adj2 = coef(fit_adj2)["inf"],
  adj3 = coef(fit_adj3)["inf"], adj4 = coef(fit_adj4)["inf"])
```

```
  adj1.inf   adj2.inf   adj3.inf   adj4.inf
0.15320864 0.13633852 0.09372376 0.12939890
```

```r
# very similar since all CIs overlap:
confint(fit_adj1, parm = "inf")
```

```
          2.5 %    97.5 %
inf 0.006778426 0.2996389
```

```r
confint(fit_adj2, parm = "inf")
```

```
          2.5 %    97.5 %
inf -0.09546615 0.3681432
```

```r
confint(fit_adj3, parm = "inf")
```

```
          2.5 %    97.5 %
inf -0.08071055 0.2681581
```

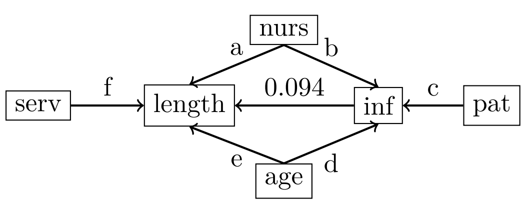
```r
confint(fit_adj4, parm = "inf")
```

```
         2.5 %    97.5 %
inf -0.1019131 0.3607109
```

e) Which adjustment set is optimal according to theory?

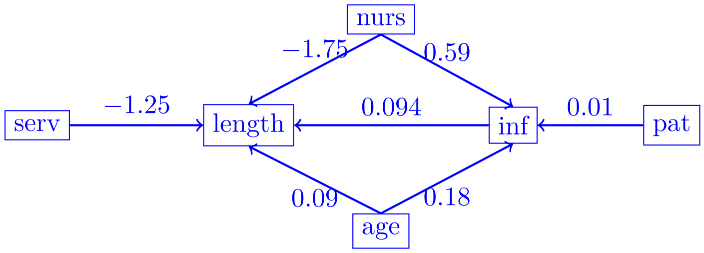
According to theory, the optimal adjustment set is pa(cn(inf,length)) without forb(inf,length).
cn(inf,length) = { length }
pa(cn(inf,length)) = { nurs, age, inf, serv }
forb(inf, length) = { inf, length }
→ Optimal adjustment set: { nurs, age, serv }
This corresponds to the regression `adj3`.
Using a suboptimal set of adjustments leads asymptotically (i.e., having a large amount of data) to greater uncertainty in the statistical conclusions. Since we do not yet have a particularly large amount of data available here, the regression adj3 does not yield the narrowest confidence intervals.

f) Calculate all causal effects (a, b, c, d, e and f) using appropriate regression models.




```r
fit1 <- lm(length ~ nurs + serv + inf + age, data = spital)
fit2 <- lm(inf ~ pat + nurs + age, data = spital)
est <- c(a = coef(fit1)["nurs"], b = coef(fit2)["nurs"],
         c = coef(fit2)["pat"], d = coef(fit2)["age"],
         e = coef(fit1)["age"], f = coef(fit1)["serv"])
round(est,2)
```

```
a.nurs b.nurs  c.pat  d.age  e.age f.serv
 -1.75   0.59   0.01   0.18   0.09  -1.25
```

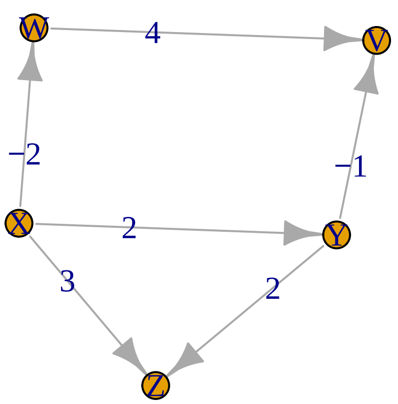



## Exercise 4: Distributions (Optional)

Suppose the three variables X, Y and Z are described by the following SCM:
$X \leftarrow E_X$
$Y \leftarrow 1 + 2 \cdot X + E_Y$
$Z \leftarrow -1 + 3 \cdot X + 2 \cdot Y + E_Z$
$W \leftarrow 3 - 2 \cdot X + E_W$
$V \leftarrow 3 - 1 \cdot Y + 4 \cdot W + E_V$
where $E_X, E_Y, E_Z, E_W, E_V \sim \mathcal{N}(0, 1)$

a) Draw a causal diagram for the given SCM.

```r
set.seed(253)
library(gRbase)
library(igraph)
gex <- dag(c("X"), c("Y","X"), c("Z","X","Y"), c("W","X"), c("V","Y","W"))
E(gex)$weight <- c(2,3,2,-2,-1,4)
plot(gex, edge.label = E(gex)$weight)
```

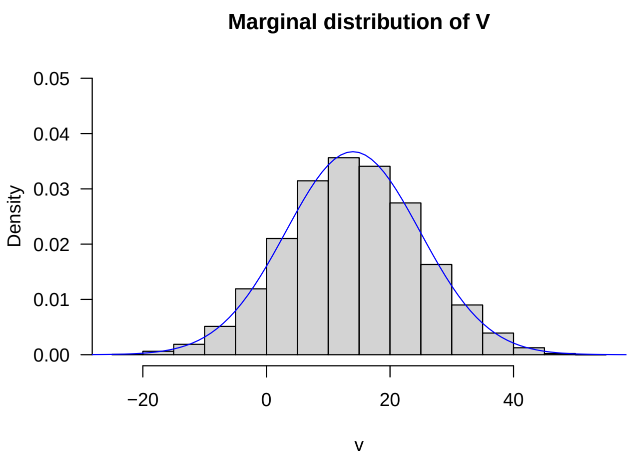



b) Simulate N = 10000 data points and visualize the distribution of V.

```r
set.seed(247)
N <- 10000
E_x <- rnorm(N)
E_y <- rnorm(N)
E_z <- rnorm(N)
E_w <- rnorm(N)
E_v <- rnorm(N)
# SCM
x <- E_x
```

```
y <- 1 + 2*x + E_y
z <- -1 + 3*x + 2*y + E_z
w <- 3 - 2*x + E_w
v <- 3 - y + 4*w + E_v
```

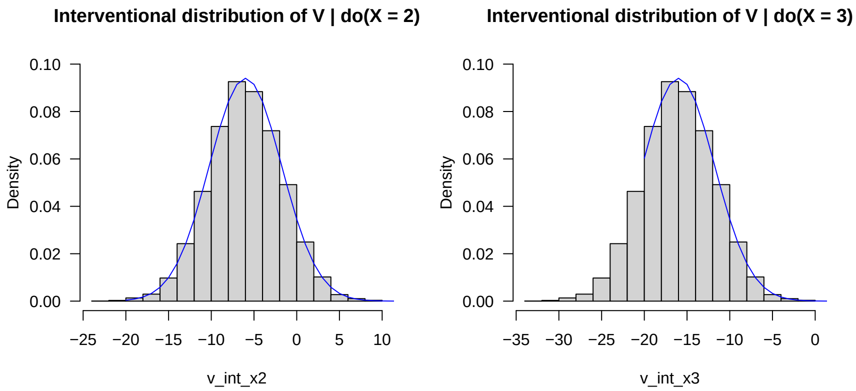
```
# Marginal distribution of V
hist(v, freq = FALSE, las = 1, ylim = c(0,0.05),
     main = "Marginal distribution of V")
lines(-30:60, dnorm(-30:60,14,sqrt(118)), col = "blue")
```




Marginal distribution of V:
$E(V) = E(3 - Y + 4 \cdot W + E_V) = 3 - E(1 + 2 \cdot X + E_Y) + 4 \cdot E(3 - 2 \cdot X + E_W) + 0 = 3 - 1 + 4 \cdot 3 = 14$
$Var(V) = Var(3 - (1 + 2 \cdot X + E_Y) + 4 \cdot (3 - 2 \cdot X + E_W) + E_V) = 10^2 \cdot Var(E_X) + Var(E_Y) + 4^2 \cdot Var(E_W) + Var(E_V) = 118$
$\Rightarrow V \sim \mathcal{N}(14, 118)$

c) Adapt the simulation of b) to compute the interventional distribution of (V | do(X = 2)) and (V | do(X = 3)).

```
# Interventions do(X = 2):
x_int2 <- rep(2,N)
y <- 1 + 2*x_int2 + E_y
w <- 3 - 2*x_int2 + E_w
v_int_x2 <- 3 - y + 4*w + E_v
# Interventions do(X = 3):
x_int3 <- rep(3,N)
y <- 1 + 2*x_int3 + E_y
w <- 3 - 2*x_int3 + E_w
v_int_x3 <- 3 - y + 4*w + E_v
par(mfrow = c(1,2))
# Interventional distribution of V | do(X = 2)
hist(v_int_x2, freq = FALSE, las = 1, ylim = c(0,0.1),
     main = "Interventional distribution of V | do(X = 2)")
lines(-20:60, dnorm(-20:60, -6, sqrt(18)), col = "blue")
# Interventional distribution of V | do(Y = 3)
hist(v_int_x3, freq = FALSE, las = 1, ylim = c(0,0.1),
     main = "Interventional distribution of V | do(X = 3)")
lines(-20:60, dnorm(-20:60, -16, sqrt(18)), col = "blue")
```




The interventional distribution of V are different since X has an indirect causal effect on V.

d) Determine the total causal effect of X on V by (1) calculating from the given SCM and (2) estimating from the simulated data (in c)).

There are two directed path from X to V:
X $\rightarrow$ W $\rightarrow$ V: $-2 \cdot 4 = -8$
X $\rightarrow$ Y $\rightarrow$ V: $2 \cdot -1 = -2$
The total causal effect is the sum of the results over both directed paths:
$-8 - 2 = -10$.

```
fit <- lm(v ~ x) # no adjustment for z needed since collider
coef(fit)
```

```
(Intercept)           x
  14.020108   -9.969687
```

e) What is the connection between c) and d)?

The shift in the expected values between the two interventional distributions is identical to the total causal effect.

f) Which of the following statements is correct?

- The marginal distribution of $Y \sim \mathcal{N}(1, 5)$
- The interventional distribution of Y | do(X = 2) is equal to the marginal distribution of Y.
- The interventional distribution of X | do(Y = 1) is equal to the marginal distribution of X.
- The interventional distribution of Y | do(X = 2) is equal to the conditional distribution of Y | X = 2.

Marginal distribution of Y:
$E(Y) = E(1 + 2 \cdot X + E_Y) = 1 + 2 \cdot E(E_X) + E(E_Y) = 1 + 0 + 0 = 1$
$Var(Y) = Var(1 + 2 \cdot X + E_Y) = 2^2 \cdot Var(E_X) + Var(E_Y) = 4 + 1 = 5$
$\Rightarrow Y \sim \mathcal{N}(1, 5)$, i.e. first statement correct.

Interventional distribution of Y | do(X = 2):
$E(Y) = E(1 + 2 \cdot 2 + E_Y) = 1 + 2 \cdot 2 + E(E_Y) = 1 + 4 + 0 = 5$
$Var(Y) = Var(1 + 2 \cdot 2 + E_Y) = Var(E_Y) = 1$
$\Rightarrow Y \sim \mathcal{N}(5, 1)$ i.e. second statement false.

Interventional distribution of X | do(Y = 1):
Since X is the cause of Y, an intervention in Y does not change the distribution of X, i.e. $X|do(Y = 1) \sim \mathcal{N}(0, 1)$.

The third statement is correct.

Conditional distribution of Y | X = 2:
Since X is direct cause of Y and the SCM has no further variables, it holds that the conditional and interventional distributions are equal.
This can also be calculated:
$Cov(X,Y) = E(XY) - E(X) \cdot E(Y) = E(E_X \cdot (1 + 2 \cdot X + E_Y)) - 0 \cdot 1 = E(E_X + 2 \cdot E_X^2 + E_X \cdot E_Y)$
$\rightarrow Cov(X,Y) = 0 + 2 + 0$ and $\rho = cor(X,Y) = Cov(X,Y)/(\sigma_X \cdot \sigma_Y) = 2/\sqrt{5}$
For a bivariate normal, the conditional distribution of $Y$ given $X = x$ is defined by
$(Y|X = x) \sim \mathcal{N}(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2))$
$\rightarrow E(Y|X = 2) = 1 + \frac{2}{\sqrt{5}} \frac{\sqrt{5}}{1}(2 - 0) = 1 + 2 \cdot 2 = 5$ and $Var(Y|X = 2) = 5 \cdot (1 - 4/5) = 1$
$\rightarrow (Y|X = 2) \sim \mathcal{N} \sim \mathcal{N}(5, 1)$
The fourth statement is correct.